

STUDENT NUMBER: _____

North Sydney Boys High School
Year 12 Physics 2022
Trial HSC Examination

General Instructions

- Reading time – 5 minutes
- Working time – 135 minutes
- Board approved calculators may be used
- Write using black pen
- Write your student number at the top of every separated page
- Use the multiple-choice answer sheet provided at the end by separating it
- A NESA data and formula sheet is provided at the end and can be separated

Total Marks – 75

Part A - 16 marks

- Attempt Questions 1–16
- Allow about 20 minutes for this part

Part B - 59 marks

- Attempt Questions 17–24
- Allow about 115 minutes for this part

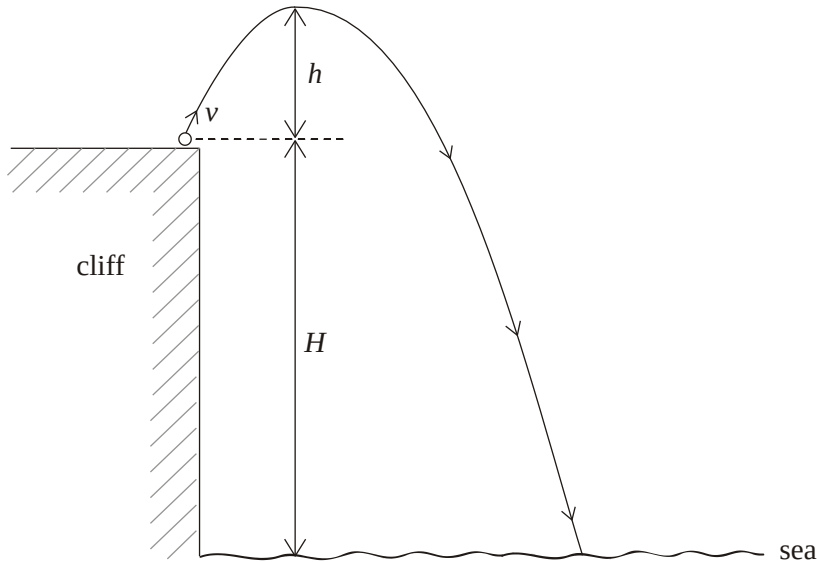
PART A /16	
PART B /59	
Total Mark /75	
KU /53	
WS /22	

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Part A – 16 marks

Choose the most correct answer. Use the multiple-choice answer sheet for Questions 1–16

1. A stone is thrown with speed v from the top of a cliff of height H , as shown below.



The stone is thrown at an angle to the horizontal so that it rises to a height h above the top of the cliff before falling into the sea. The acceleration of free fall is g . Air resistance is negligible.

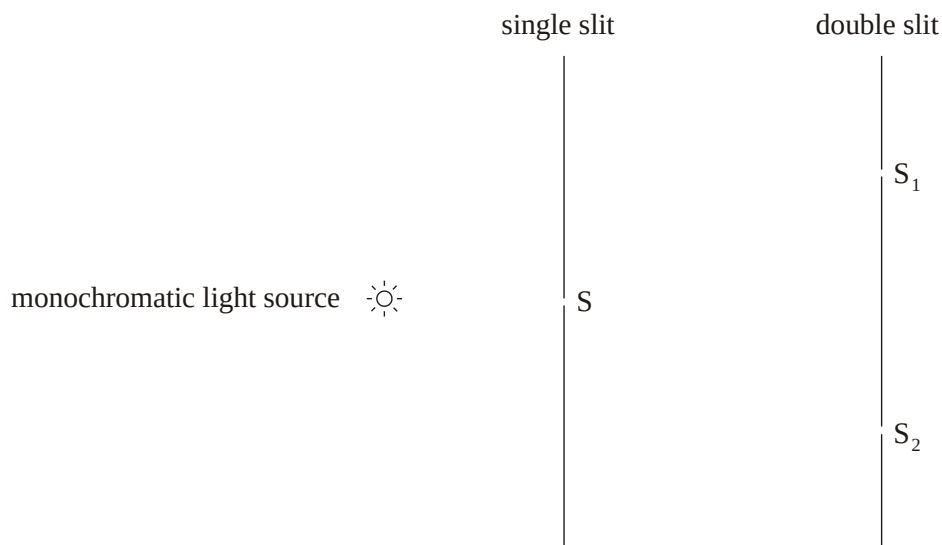
Which **one** of the following expressions correctly gives the speed of the stone as it hits the sea?

- A. $v + \sqrt{2gh}$
 B. $v + \sqrt{2gH}$
 C. $\sqrt{2g\{h+H\}}$
 D. $\sqrt{v^2 + 2gH}$
2. A transformer has a primary coil with N_p turns and a secondary coil with N_s turns. An alternating voltage supply of frequency f and voltage V_p is connected to the primary coil.

Which of the following correctly gives the frequency and voltage in the secondary coil?

	Frequency	Voltage
A.	$\frac{N_s}{N_p} f$	$\frac{N_p}{N_s} V_p$
B.	f	$\frac{N_p}{N_s} V_p$
C.	$\frac{N_p}{N_s} f$	$\frac{N_s}{N_p} V_p$
D.	f	$\frac{N_s}{N_p} V_p$

3. The rings of Saturn are made of rocky particles that orbit the planet. The period T of each particle depends on its distance r from the centre of Saturn. The period T is proportional to r^n . Which **one** of the following is n equal to?
- A. 1.0
- B. 1.5
- C. 2.0
- D. 3.0
4. The diagram below shows the arrangement for a Young's double slit experiment.



The function of the single slit is

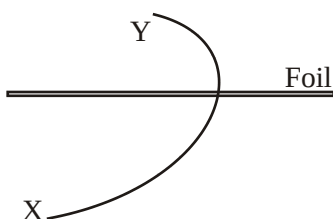
- A. to direct the light towards S_1 and S_2 .
- B. to ensure equal intensities of light at S_1 and S_2 .
- C. to produce coherent light at S_1 and S_2 .
- D. to reduce the intensity of light at S_1 and S_2 .
5. A proton of mass m and charge e is accelerated from rest through a potential difference V . The final speed of the proton is
- A. $\sqrt{\frac{2Ve}{m}}$.
- B. $\frac{2Ve}{m}$.
- C. $\sqrt{\frac{Ve}{m}}$.
- D. $\frac{Ve}{m}$.

6. Newton's law of gravitation for the force F between two point objects of masses M and m , separated by a distance d may be written as

$$Fd^2 \propto Mm.$$

The expression may also be used for the force of attraction between the Sun and the Earth, although they are not point masses. This is because

- A. the gravitational constant G is not involved in the expression.
 B. the force between the Sun and the Earth is very large.
 C. the separation of the Sun and the Earth is much greater than their radii.
 D. the mass of the Earth is much less than the mass of the Sun.
7. The diagram below shows the path XY of an electron that passes through a thin foil placed in a vacuum.



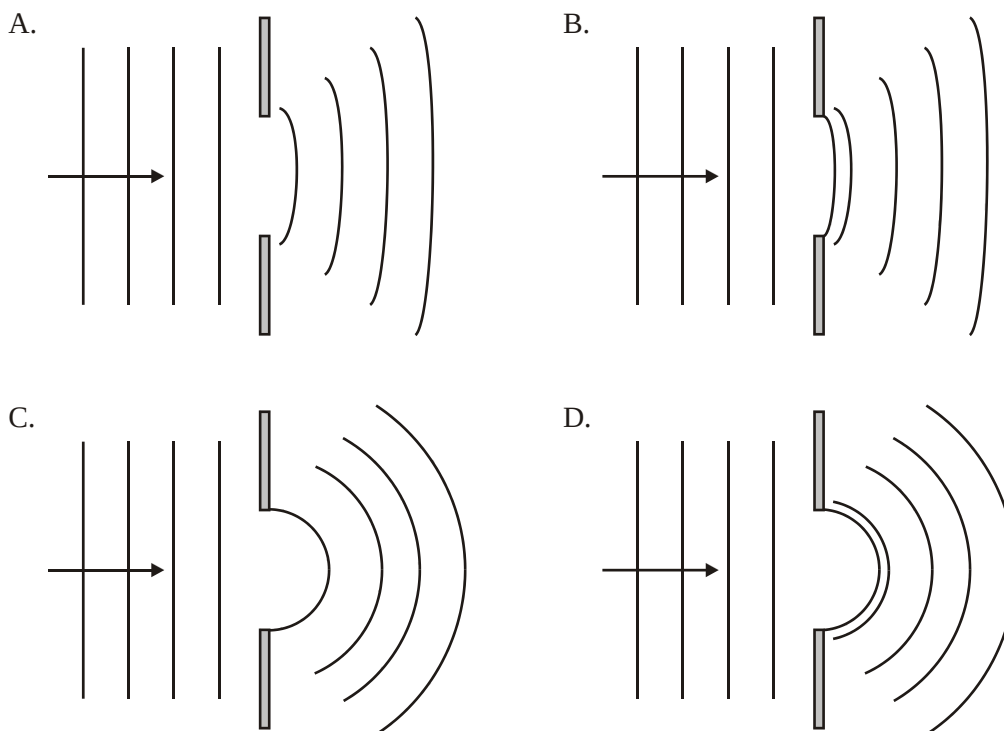
There is a uniform magnetic field acting perpendicularly to the plane of the page. Which **one** of the following statements best explains the path followed by the electron?

	Direction of motion of the electron	Direction of the magnetic field
A.	Y to X	into the plane of the page
B.	Y to X	out of the plane of the page
C.	X to Y	into the plane of the page
D.	X to Y	out of the plane of the page

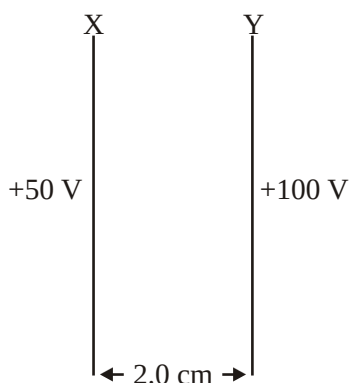
8. A satellite is in orbit about Earth. The satellite moves to an orbit closer to Earth. Which of the following correctly gives the change in the potential energy and the kinetic energy of the satellite?

	change in potential energy	change in kinetic energy
A.	Decreases	Increases
B.	Decreases	Decreases
C.	Increases	Increases
D.	Increases	Decreases

9. Which diagram best shows diffraction of plane wavefronts at a single slit?



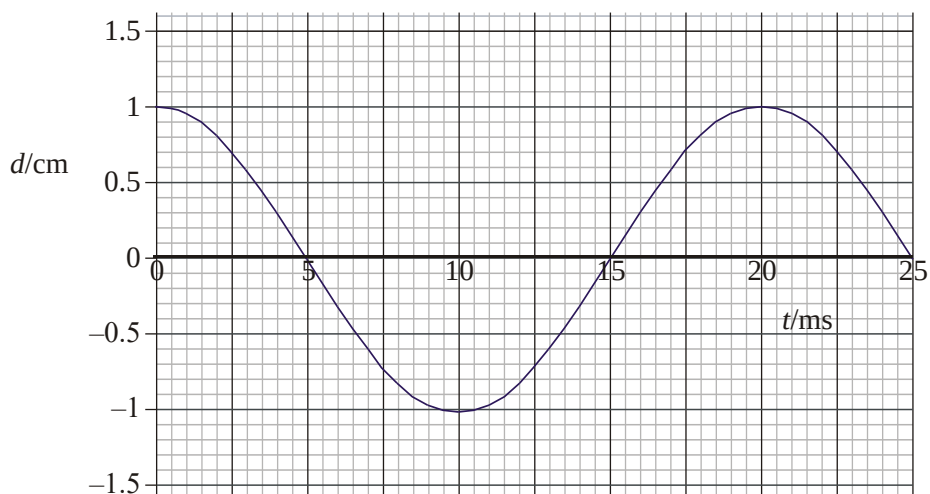
10. The diagram below shows two lines of equipotential in a region of a uniform electric field. Line X has a potential of +50 V and line Y has a potential of +100V. The distance between X and Y is 2.0 cm.



Which **one** of the following correctly gives the direction of the electric field and its strength?

	Direction	Strength / V cm^{-1}
A.	$X \rightarrow Y$	25
B.	$X \rightarrow Y$	100
C.	$Y \rightarrow X$	25
D.	$Y \rightarrow X$	100

11. Two satellites of equal mass, S_1 and S_2 , orbit the Earth. S_1 is orbiting at a distance r from the Earth's centre at speed v . S_2 orbits at a distance $2r$ from the Earth's centre at speed $\frac{v}{\sqrt{2}}$. The ratio of the centripetal force on S_1 to the centripetal force on S_2 is
- A. $\frac{1}{8}$.
- B. $\frac{1}{4}$.
- C. 4.
- D. 8.
12. A wave is travelling through a medium. The diagram shows the variation with time t of the displacement d of a particle of the medium from $t = 0$ to $t = 25$ ms.



Which of the following correctly gives the frequency and the amplitude of the wave?

	frequency / Hz	amplitude / cm
A.	2.0×10^{-2}	2.0
B.	2.0×10^{-2}	1.0
C.	50	2.0
D.	50	1.0

13. Which **one** of the following is a correct definition of electric potential difference between two points?
- A. The power to move a small positive charge between the two points.
 - B. The work done to move a small positive charge between the two points.
 - C. The power per unit charge to move a small positive charge between the two points.
 - D. The work done per unit charge to move a small positive charge between the two points.
14. The acceleration of free fall of a small sphere of mass $5.0 \times 10^{-3} \text{ kg}$ when close to the surface of Jupiter is 25 ms^{-2} . The gravitational field strength at the surface of Jupiter is
- A. $2.0 \times 10^{-4} \text{ N kg}^{-1}$.
 - B. $1.3 \times 10^{-1} \text{ N kg}^{-1}$.
 - C. 25 N kg^{-1} .
 - D. $5.0 \times 10^3 \text{ N kg}^{-1}$.
15. The waves from two light sources meet at a point. Which condition is essential for interference to be observed?
- A. Constant phase difference between the waves
 - B. Equal amplitude of the waves
 - C. Equal frequency of the waves
 - D. Equal intensities of the waves
16. The escape speed of an object of mass m from a planet of mass M and radius r depends on the gravitational constant and
- A. M and r .
 - B. m and r .
 - C. M only.
 - D. M , m , and r .

Part B – 59 marks

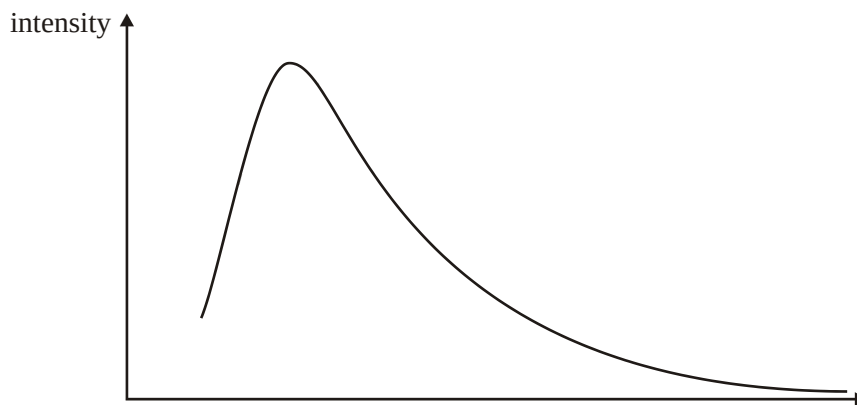
Write your answers in the spaces provided

17. (a) Explain the term *black-body radiation*.

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(1)

The diagram below is a sketch graph of the black-body radiation spectrum of a certain star.



- (b) Label the x -axis of the graph.

(1)

- (c) On the graph, sketch the black-body radiation spectrum of a star that has a lower surface temperature and lower apparent brightness than this star.

(1)

The star Betelgeuse in the Orion constellation emits black-body radiation that has a maximum intensity at a wavelength of $0.97\ \mu\text{m}$.

- (d) Deduce that the surface temperature of Betelgeuse is about 3000 K.

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(1)

(Total 4 marks)

18. (a) State what is meant by polarised light.

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(1)

- (b) Polarised light of intensity I_0 is incident on an analyser. The transmission axis of the analyser makes an angle θ with the direction of the electric field of the light.

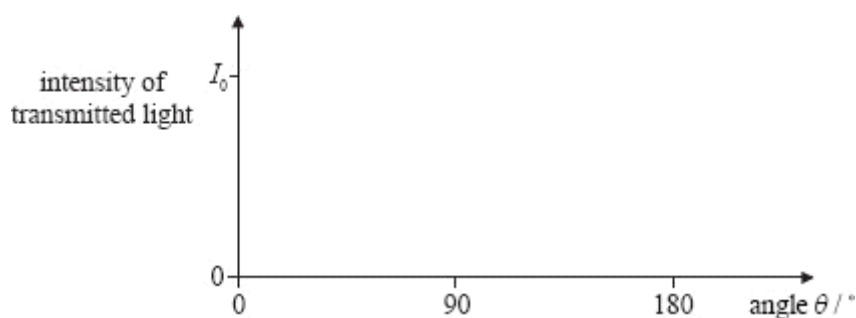
- (i) Calculate, in terms of I_0 , the intensity of light transmitted through the analyser when $\theta = 60^\circ$.

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(1)

- (ii) On the axes below, sketch a graph to show the variation with angle θ of the intensity of the transmitted light.



(2)

- (c) Outline how polarising sunglasses reduce glare from a reflecting surface.

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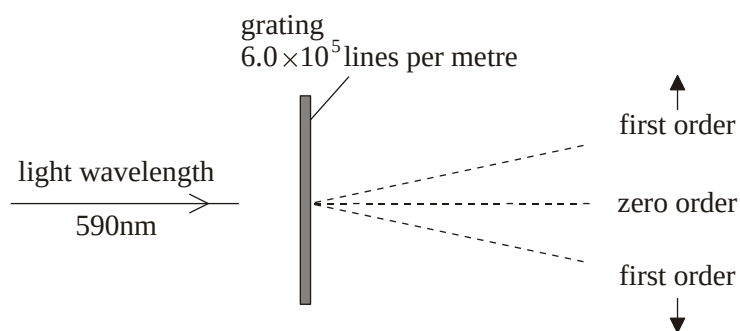
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(2)

(Total 6 marks)

19. Light of wavelength 590 nm is incident normally on a diffraction grating, as shown below.



The grating has 6.0×10^5 lines per metre.

- (a) Determine the **total** number of orders (both sides) of diffracted light, including the zero order, that can be observed.

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(4)

- (b) The incident light is replaced by a beam of light consisting of two wavelengths, 590 nm and 589 nm.

State **two** observable differences between a first order spectrum and a second order spectrum of the diffracted light.

1.
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2.
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(2)

(Total 6 marks)

20. (a) A satellite goes round a planet at constant speed. Explain why, although its speed is constant, it is accelerating.

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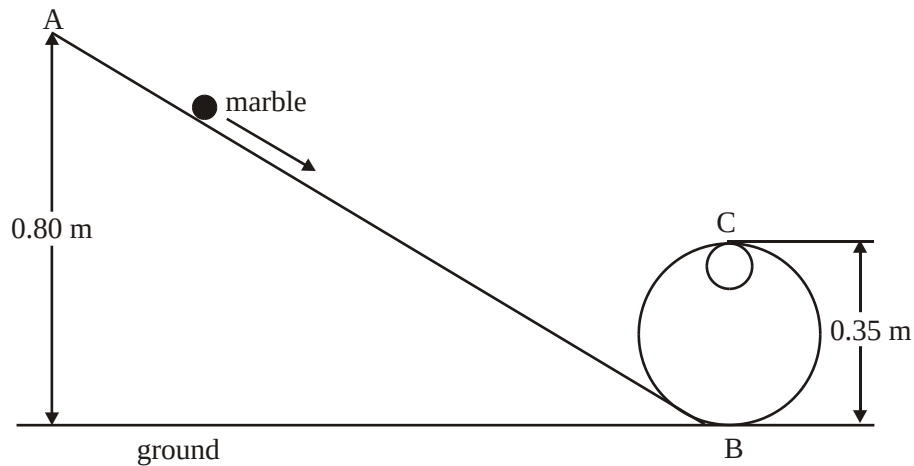
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(1)

In the diagram below, a marble (small glass sphere) rolls down a track, the bottom part of which has been bent into a loop. The end A of the track, from which the marble is released, is at a height of 0.80 m above the ground. Point B is the lowest point and point C the highest point of the loop. The diameter of the loop is 0.35 m.



The mass of the marble is 0.050 kg. Friction forces and any gain in kinetic energy due to the rotating of the marble can be ignored.

Consider the marble when it is at point C.

- (b) (i) On the diagram above, draw an arrow to show the direction of the resultant force acting on the marble.

(1)

- (ii) State the names of the **two** forces acting on the marble.

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(2)

- (iii) Deduce that the speed of the marble is approximately 3.0 ms^{-1} .

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(2)

- (iv) Determine the resultant force acting on the marble and hence determine the reaction force of the track on the marble.

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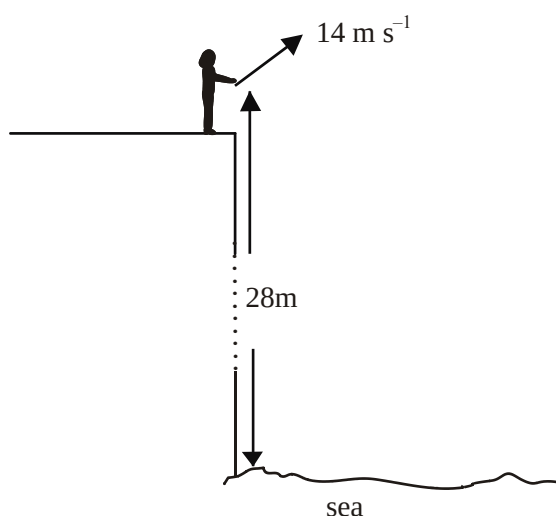
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(3)

(Total 9 marks)

21. A stone is thrown from the top of a cliff of height 28 m above the sea. The stone is thrown at a speed of 14 m s^{-1} at an angle above the horizontal. Air resistance is negligible.



The maximum height reached by the stone measured from the point from which it is thrown is 8.0 m.

- (a) By considering the energy of the stone, determine the speed with which the stone hits the sea.

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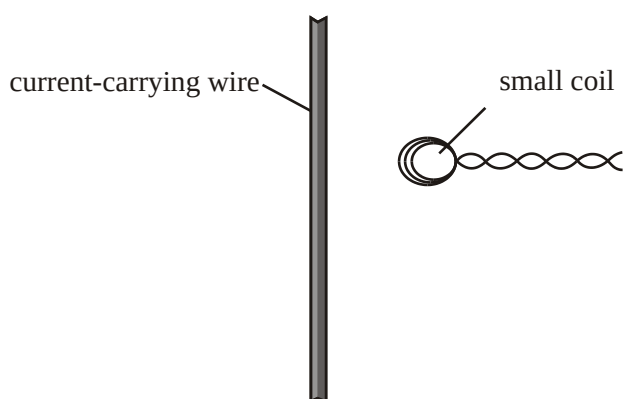
(2)

- (b) The stone leaves the cliff at time $t = 0$. It reaches its maximum height at $t = T_H$. On the axis below, draw a sketch-graph to show the variation with time t of the magnitude of the **vertical** component of speed of the stone from $t = 0$ to $t = T_S$, the time just before the stone strikes the sea.



(2)
(Total 4 marks)

22. A small coil is placed with its plane parallel to a long straight current-carrying wire, as shown below.



- (a) (i) State Faraday's law of electromagnetic induction.

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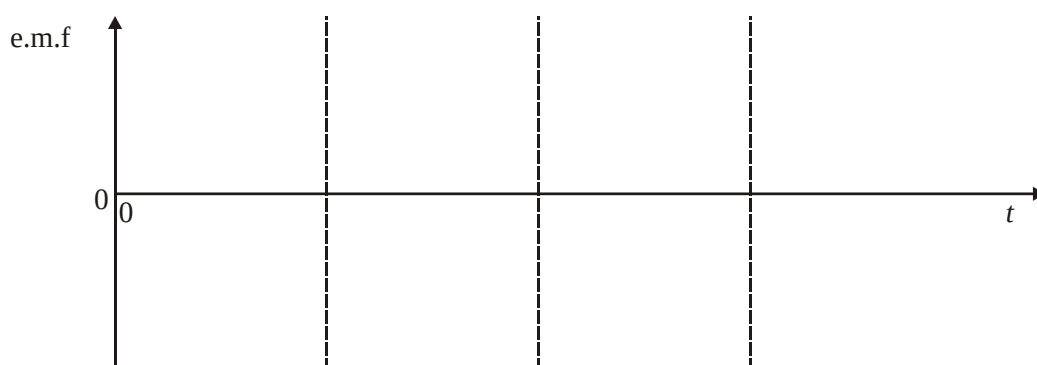
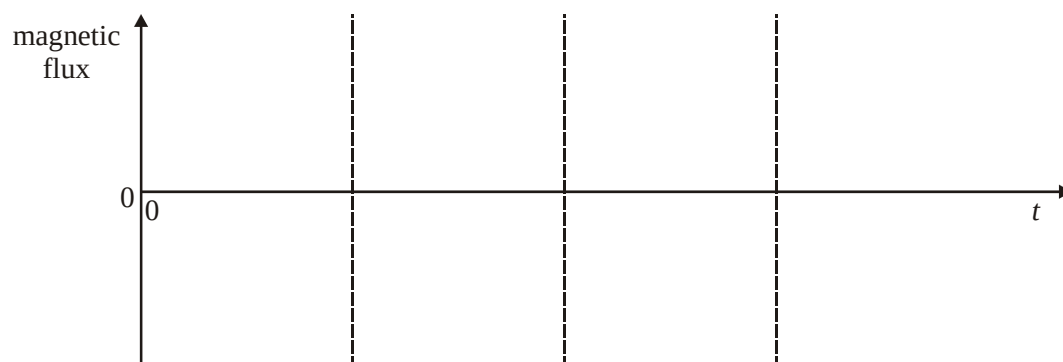
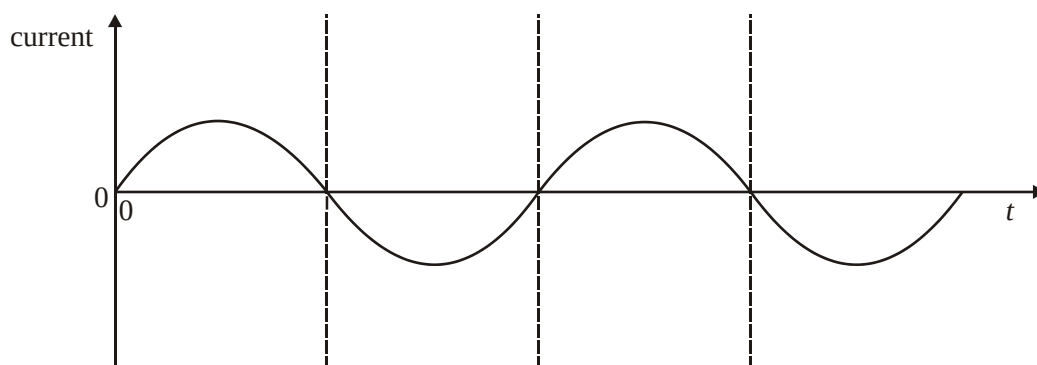
(2)

- (ii) Use the law to explain why, when the current in the wire changes, an emf is induced in the coil.

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(1)

The diagram below shows the variation with time t of the current in the wire.



- (b) (i) Draw, on the axes provided, a sketch-graph to show the variation with time t of the magnetic flux in the coil. (1)
- (ii) Construct, on the axes provided, a sketch-graph to show the variation with time t of the emf induced in the coil. (2)
- (iii) State and explain the effect on the maximum emf induced in the coil when the coil is further away from the wire.

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(2)

- (c) Such a coil may be used to measure large alternating currents in a high-voltage cable. Identify **one** advantage and **one** disadvantage of this method.

Advantage:

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Disadvantage:

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(2)
(Total 10 marks)

23. This question is about Kepler's third law.

- (a) Kepler's third law states that the period T of the orbit of a planet about the Sun is related to the average orbital radius R of the planet by the relationship

$$T^2 = KR^3$$

where K is a constant.

- (i) Suggest why the law specifies the average orbital radius.

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(1)

- (ii) State the name of the force that causes the acceleration of the planets orbiting the Sun.

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(1)

- (iii) State an expression for the magnitude F of the force in (ii) in terms of the mass M_S , of the Sun, the mass m of the planet, the radius R of the orbit and the universal gravitational constant G .

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(1)

- (iv) Hence deduce, explaining your working, that the constant K is given by the expression

$$K = \frac{4\pi^2}{GM_S}$$

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(4)

- (b) Ganymede is one of the moons of Jupiter and the following data are available.

Average orbital radius of Ganymede = 1.1×10^9 m

Orbital period of Ganymede = 6.2×10^5 s

Universal gravitational constant $G = 6.67 \times 10^{-11}$ N m² kg⁻²

- (i) Deduce that the gravitational field strength of Jupiter at the surface of Ganymede is approximately 0.1 N kg^{-1} .

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(2)

- (ii) Estimate the mass of Jupiter.

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(3)

(Total 12 marks)

24. (a) Describe one historical method used to determine the speed of light experimentally including its limitations.

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- (b) In 1983, the speed of light was declared to be exactly $299\,792\,458\text{ ms}^{-1}$. Explain the currently accepted relationship of the speed of light to the measurement of distance and time.

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(2)
(Total 8 marks)

End of paper

DATA SHEET

FORMULAE SHEET

		Motion, forces and gravity
Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$	$s = ut + \frac{1}{2}at^2$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$	$\vec{F}_{\text{net}} = m\vec{a}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$	$W = F_{\parallel}s = F_s \cos \theta$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$	$K = \frac{1}{2}mv^2$
Speed of sound in air	340 m s^{-1}	$P = F_{\parallel}v = F_v \cos \theta$
Earth's gravitational acceleration, g	9.8 m s^{-2}	$\sum \frac{1}{2}mv^2_{\text{before}} = \sum \frac{1}{2}mv^2_{\text{after}}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$	$a_c = \frac{v^2}{r}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$	$F_c = \frac{mv^2}{r}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$	$F = \frac{GMm}{r^2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$	$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$	
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$	
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$	
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$	$f_{\text{beat}} = f_2 - f_1 $
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$	$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$
1 eV	$931.5 \text{ MeV}/c^2$	$n_1 \sin \theta_1 = n_2 \sin \theta_2$
Density of water, ρ	$1.602 \times 10^{-19} \text{ J}$	$\sin \theta_c = \frac{n_2}{n_1}$
Specific heat capacity of water	$1.00 \times 10^3 \text{ kg m}^{-3}$	$I_1 r_1^2 = I_2 r_2^2$
Wien's displacement constant, b	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$	$\frac{Q}{t} = \frac{kA\Delta T}{d}$
	$2.898 \times 10^{-3} \text{ m K}$	
		Waves and thermodynamics
		$v = f\lambda$
		$f = \frac{1}{T}$
		$d \sin \theta = m\lambda$
		$n_x = \frac{c}{v_x}$
		$I = I_{\text{max}} \cos^2 \theta$
		$Q = mc\Delta T$

Electricity and magnetism

$$E = \frac{V}{d}$$
$$V = \frac{\Delta U}{d}$$
$$W = qV$$
$$W = qEd$$
$$B = \frac{\mu_0 I}{2\pi r}$$
$$B = \frac{\mu_0 NI}{L}$$
$$\Phi = B_{\parallel}A = BA\cos\theta$$
$$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$$
$$V_p = \frac{N_p}{N_s}$$
$$V_s = \frac{N_s}{N_p}$$

$$\vec{F} = q\vec{E}$$
$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2}$$
$$I = \frac{q}{t}$$
$$V = IR$$
$$P = VI$$
$$F = q\vec{v} \cdot \vec{B} = qvB\sin\theta$$
$$F = I\vec{L} \cdot \vec{B} = IIB\sin\theta$$
$$F = \frac{\mu_0}{2\pi} \frac{I_1I_2}{r}$$
$$\tau = nI\vec{A} \cdot \vec{B} = nIAB\sin\theta$$
$$V_d = \frac{VI_s}{s}$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$
$$K_{\max} = hf - \phi$$
$$\lambda_{\max} = \frac{b}{T}$$
$$E = mc^2$$
$$E = hf$$
$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$t = \frac{t_0}{\sqrt{1-\frac{v^2}{c^2}}}$$
$$l = l_0\sqrt{1-\frac{v^2}{c^2}}$$
$$p_v = \frac{m_0v}{\sqrt{1-\frac{v^2}{c^2}}}$$
$$N = N_0e^{-\lambda t}$$
$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$$

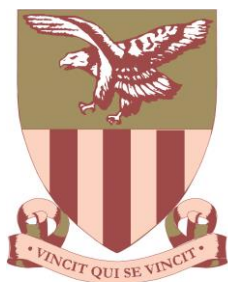
PERIODIC TABLE OF THE ELEMENTS

2	He	4.003	Helium
3	H	1.008	Hydrogen
4	Be	9.012	Lithium
5	B	10.81	Boron
6	C	12.01	Carbon
7	N	14.01	Nitrogen
8	O	16.00	Oxygen
9	F	19.00	Fluorine
10	Ne	20.18	Neon
11	Na	22.99	Sodium
12	Mg	24.31	Magnesium
13	Al	26.98	Aluminium
14	Si	28.09	Silicon
15	P	30.97	Phosphorus
16	S	32.07	Sulfur
17	Cl	35.45	Chlorine
18	Ar	39.95	Argon
19	K	39.10	Potassium
20	Ca	40.08	Calcium
21	Sc	44.96	Scandium
22	Ti	47.87	Titanium
23	V	50.94	Vanadium
24	Cr	52.00	Chromium
25	Mn	54.94	Manganese
26	Fe	55.85	Iron
27	Co	58.93	Cobalt
28	Ni	58.69	Nickel
29	Cu	63.55	Copper
30	Zn	65.38	Zinc
31	Ga	69.72	Gallium
32	Ge	72.64	Germanium
33	As	74.92	Arsenic
34	Se	78.96	Selenium
35	Br	79.90	Bromine
36	Kr	83.80	Krypton
37	Rb	85.47	Rubidium
38	Sr	87.61	Strontium
39	Y	88.91	Yttrium
40	Zr	91.22	Zirconium
41	Nb	92.91	Niobium
42	Mo	95.96	Molybdenum
43	Tc		Technetium
44	Ru	101.1	Ruthenium
45	Rh	102.9	Rhodium
46	Pd	106.4	Palladium
47	Ag	107.9	Silver
48	Cd	112.4	Cadmium
49	In	114.8	Indium
50	Sn	118.7	Tin
51	Sb	121.8	Antimony
52	Te	127.6	Tellurium
53	I	126.9	Iodine
54	Xe	131.3	Xenon
55	Cs	132.9	Cesium
56	Ba	137.3	Barium
57-71			Lanthanoids
58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium
61	Pm		Promethium
62	Sm	150.4	Samarium
63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium
68	Er	167.3	Erbium
69	Tm	168.9	Thulium
70	Yb	173.1	Ytterbium
71	Lu	175.0	Lutetium
72	Hf	178.5	Hafnium
73	Ta	180.9	Tantalum
74	W	183.9	Tungsten
75	Re	186.2	Rhenium
76	Os	190.2	Osmium
77	Ir	192.2	Iridium
78	Pt	195.1	Platinum
79	Au	197.0	Gold
80	Hg	200.6	Mercury
81	Tl	204.4	Thallium
82	Pb	207.2	Lead
83	Bi	209.0	Bismuth
84	Po		Polonium
85	At		Astatine
86	Rn		Radon
87	Fr		Francium
88	Ra		Radium
89-103			Actinoids
104	Rf		Rutherfordium
105	Db		Dubnium
106	Sg		Seaborgium
107	Bh		Bohrium
108	Hs		Hassium
109	Mt		Meitnerium
110	Ds		Darmstadtium
111	Rg		Roentgenium
112	Cn		Copernicium
113	Nh		Nihonium
114	Fl		Flerovium
115	Mc		Moscovium
116	Lv		Livermorium
117	Ts		Tennesine
118	Og		Oganesson

Atomic Number	Symbol	Standard Atomic Weight	Name
79	Au	197.0	Gold

KEY

STUDENT NUMBER: _____



North Sydney Boys High School

Science Faculty

HSC Physics 2022

Assessment 3

MULTIPLE CHOICE ANSWER SHEET

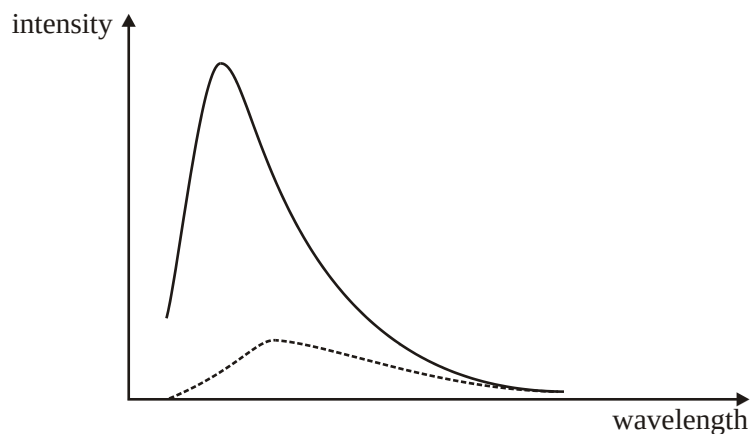
- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 11. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 12. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 13. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 14. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 15. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 16. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Marking Guideline Year 12 Physics Trial Exam 2022

Note: Working Scientifically marks have (WS)

- | | | |
|-----|---|---|
| 1. | D | 1 |
| 2. | D | |
| 3. | B | |
| 4. | C | |
| 5. | A | |
| 6. | C | |
| 7. | D | |
| 8. | A | |
| 9. | A | |
| 10. | C | |
| 11. | C | |
| 12. | D | |
| 13. | D | |
| 14. | C | |
| 15. | A | |
| 16. | A | |

- | | | |
|-----|--|---|
| 17. | (a) the radiation emitted by a perfect emitter or a perfect absorber | 1 |
| | (b) wavelength / λ ; | 1 |
| | (c) | |



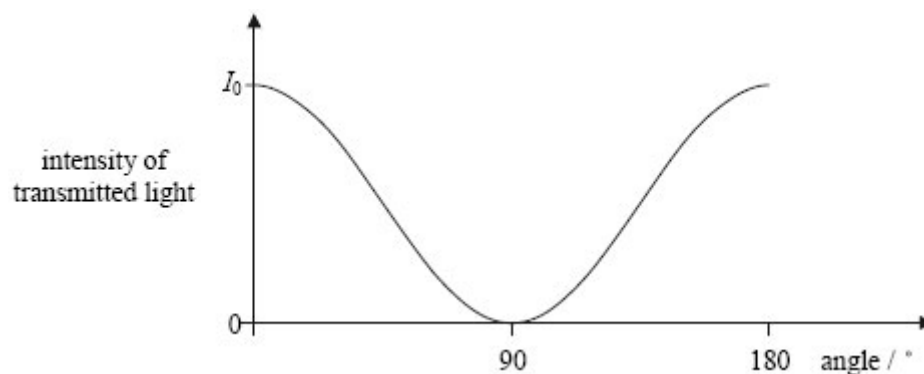
lower intensities with peak shifted to the longer wavelength;

(d)	$T = \frac{2.90 \times 10^{-3}}{\lambda} = \frac{2.90 \times 10^{-3}}{9.70 \times 10^{-7}} = 3000 \text{ K};$	1
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[4]

- | | | |
|-----|---|---|
| 18. | (a) light where the direction of the (electric) field is always (or predominantly) in the same plane; | 1 |
| | (b) (i) $I = (I_0 \cos^2 60^\circ) = \frac{I_0}{4};$ | 1 |

(ii)



general $\cos^2$ shape;

1 (WS)

max at $\theta = 0$ and curve touches horizontal axis at $\theta = 90^\circ$;

1 (WS)

- (c) light is (partially) horizontally polarised by reflection;
sunglasses have a transmission axis at 90° to the plane of reflected light
causing the intensity of reflected light to be reduced;

1

1

[6]

19. (a) using $d \sin \theta = m\lambda$ and the idea that max m is achieved when $\sin \theta = 1$;

1

making use of $d = \frac{1}{6.0 \times 10^5}$;

1

to give $m = 2.8$;

1

hence there are five complete orders; (2 orders on each side including zeroth)

1

- (b) Spectrum/lines are brighter in the first order;
Lines more likely to be seen separately in second order;

1

1

[6]

20. (a) the direction of the planet's trajectory is changing hence the velocity
of the planet is changing;

1

- (b) (i) arrow pointing vertically downwards;

1

- (ii) weight (or gravity);
normal force/reaction force;

1

1

- (iii) loss in PE = $0.05 \times 9.8 \times (0.8 - 0.35)$;
= gain in KE = $\frac{1}{2}mv^2$;

1

to give $v = 2.97 \text{ m s}^{-1}$ (3.0 m s^{-1});

1

or

use of $v = \sqrt{2gh}$ to give $v = 4.0 \text{ m s}^{-1}$ at point B;

and then use of $v^2 - u^2 = 2gh$ with $u = 4.0 \text{ m s}^{-1}$ and $h = 0.35 \text{ m}$;
 $g = -9.8 \text{ m s}^{-2}$ to get $v = 3.0 \text{ m s}^{-1}$;

- (iv) recognise that resultant force = $\frac{mv^2}{r} = \frac{(0.05 \times 9.0)}{0.175} = 2.6 \text{ N}$;

1

$N = \frac{mv^2}{r} - mg$; (must be stated)

1

$= 2.6 - 0.5 = 2.1 \text{ N}$;

1

[9]

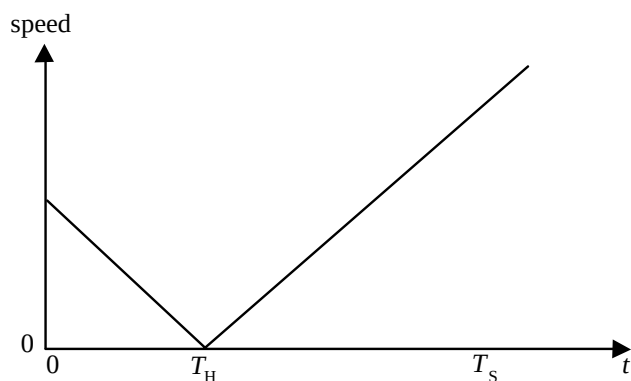
21. (a) initial KE of stone + loss in PE = gain in KE (must indicate this relationship);

1

$\frac{14^2}{2} + 9.8 \times 28 = \frac{v^2}{2}$ to give $v = 27 \text{ m s}^{-1}$ (mass cancels out);

1

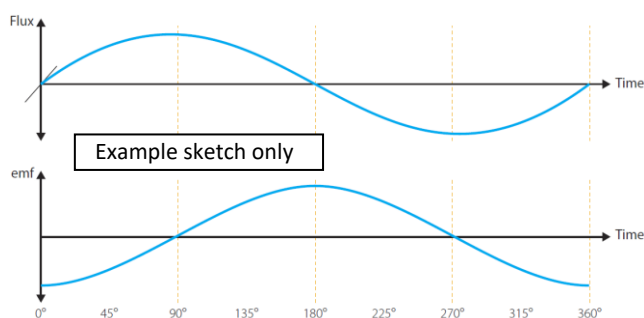
(b)



two lines going in the correct direction as shown; 1 (WS)
 speed greater at T_S than initial speed and zero at T_H (slope magnitudes equal); 1 (WS)

[4]

22. (a) (i) emf (induced) is proportional to; 1 (WS)
 rate of change / cutting of (magnetic) flux; 1 (WS)
- (ii) magnetic field / flux through coil will change as the current changes; 1 (WS)
- (b) (i) sinusoidal and in phase with current; 1 (WS)
 (ii) sinusoidal and same frequency; 1 (WS)
 with 90° phase difference to graph for ϕ and Lenz's law implemented; 1 (WS)



- (iii) emf is reduced; 1 (WS)
 because B is smaller; 1 (WS)
- (c) advantage: no direct contact with cable required; 1 (WS)
 disadvantage: distance to wire must be fixed; 1 (WS)

[10]

23. Kepler's third law

- (a) (i) the orbits are elliptical / not circular; 1
Do not accept "because R changes".
- (ii) gravity / gravitational; 1
- (iii) $F = G \frac{M_s m}{R^2}$; 1

Same symbols as in question must be used to receive the mark.

- (iv) let v = the speed of planet, the acceleration is then = $\frac{v^2}{R}$ 1

$$\text{from Newton 2, } F = G \frac{M_s m}{R^2} = \frac{mv^2}{R}; \left\{ \begin{array}{l} \text{Some reference to Newton 2} \\ \text{is required to receive this mark.} \end{array} \right. \quad 1$$

$$v = \frac{2\pi R}{T}; \text{ therefore, } = G \frac{M_s m}{R^2} = \frac{4\pi^2 m R^2}{T^2}; \quad 1$$

$$\text{therefore, } T^2 = \frac{4\pi^2}{GM_s} R^3; \text{ so } K = \frac{4\pi^2}{GM_s} \quad 1$$

(b) (i) recognise that gravitational field strength = $\frac{v^2}{R} = \frac{4\pi^2 R}{T^2}; \quad 1$

$$= \frac{40 \times 1.1 \times 10^9}{(6.2 \times 10^5)^2}; = 0.1 \text{ N kg}^{-1} \quad 1$$

(ii) $T^2 = \frac{4\pi^2}{GM_s} R^3$ therefore, $M_{jup} = \frac{4\pi^2 R^3}{GT^2}; \quad 2$

$$M_{jup} = \frac{4\pi^2 \times (1.1)^3 \times 10^{27}}{6.7 \times 10^{-11} \times (6.2 \times 10^5)^2}; \approx 2.0 \times 10^{27} \text{ kg}; \quad 1$$

Or

$$0.1 = \frac{GM}{R^2}; M = \frac{0.1R^2}{G}; = \frac{0.1 \times (1.1 \times 10^9)^2}{6.7 \times 10^{-11}} \approx 2.0 \times 10^{27} \text{ kg};$$

[12]

24.

Historical method to measure speed of light experimentally	Marks
Provides a basic, labelled sketch of the procedure used	1(WS)
Describes the basic operation of the method. (Determination of speed purely from theory such as from Maxwell's laws is not accepted)	1(WS)
Explains the physics principles behind the method (example: reflection of light, law of reflection, finite speed of light, orbital motion of planets/moons, eclipses, light aberration, interference patterns, angular rotation/speed, etc.) Need to list two physics principles at least.	2(WS)
Describes two limitations of the technique/results obtained (example: slow human reaction time, short distances for light travel, limitation of timing devices, irregular orbits, etc)	2(WS)

Relationship of speed of light to time and distance	Marks
Explains that the metre is defined as the length of the path travelled by light in a vacuum in 1/c seconds	1(WS)
Explains that one second is defined as the time taken for a fixed number (9,192,631,770) of energy transitions in a Caesium 133 atom. (Note: exact number of transitions are not required but Caesium 133 atom must be stated)	1(WS)

[8]